

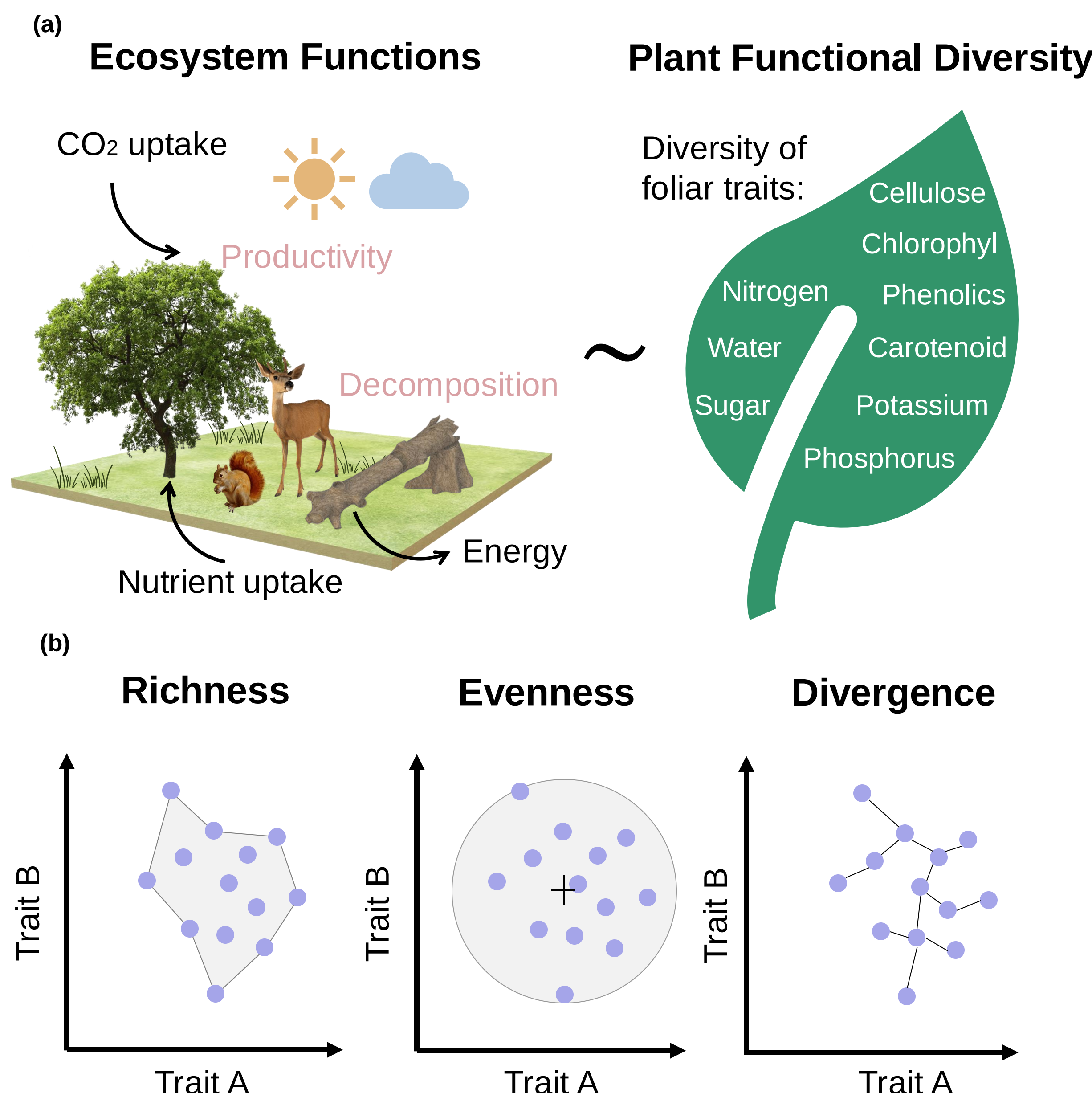
## Introduction

Plant functional diversity is a key indicator of various ecosystem functions, including primary production, plant health, water cycles, decomposition, and ecosystem stability. Higher functional diversity has been documented to facilitate resource and niche partitioning, leading to enhanced ecosystem functions such as accelerated carbon cycles and more effective nutrient transition within the food web. However, studies on the connection between functional diversity and ecosystem functions (FD-EF) were usually restricted to small scales due to the high costs and low efficiencies in data collection. Leveraging evidence from imaging spectroscopy, this study aim to understand how the FD-EF relationship varies across different biomes in North America.

## Objectives

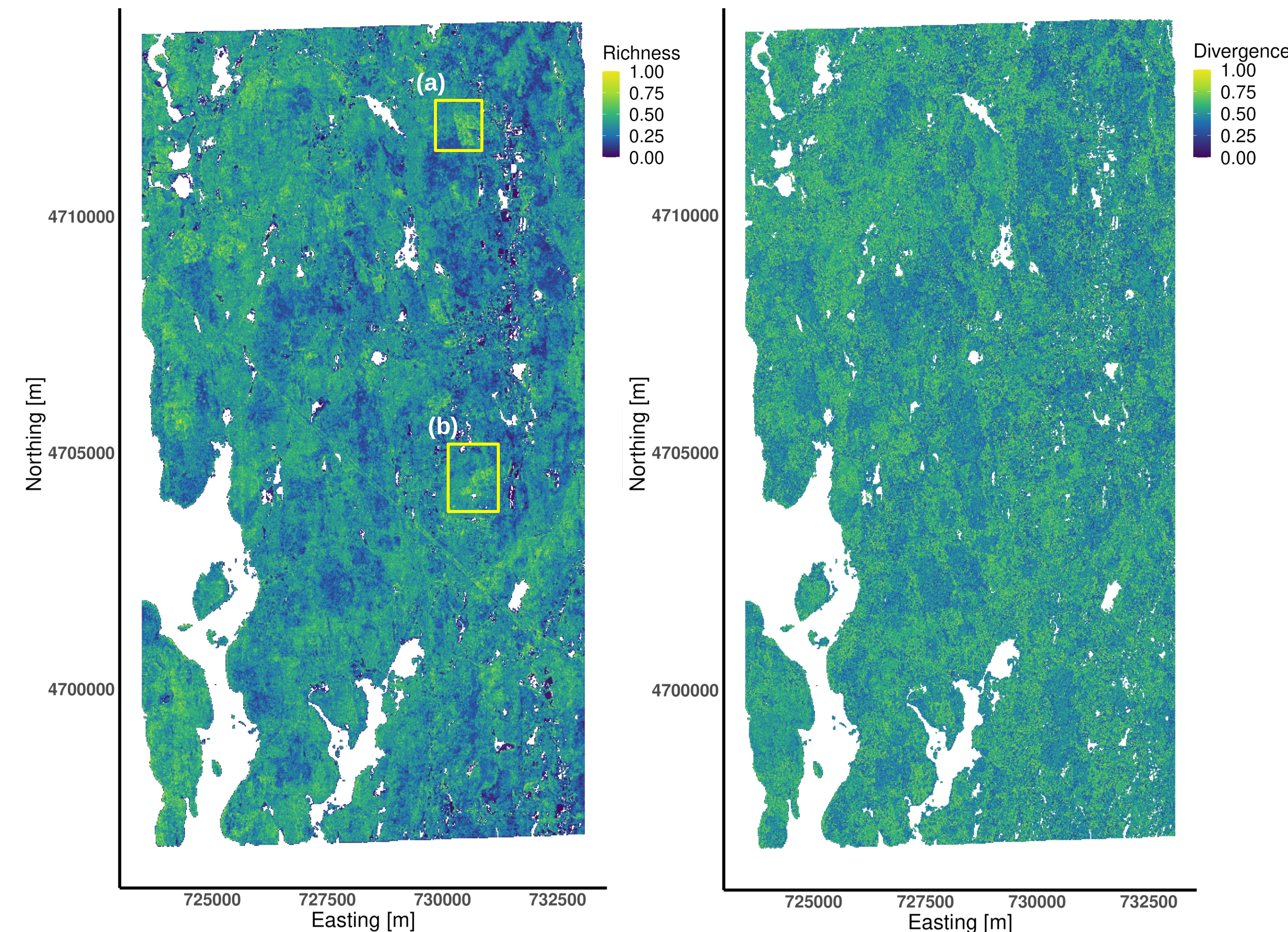
- We derived multidimensional functional diversity from remotely sensed functional trait maps, including functional richness, evenness, and divergence, across 39 terrestrial sites within the National Ecological Observatory Network (NEON).
- We integrated NEON terrestrial data with airborne and satellite remote sensing products to quantify ecosystem functions, using Gross Primary Production (GPP) and Leaf Area Index (LAI) for productivity, as well as Coarse Woody Debris (CWD) for decomposition.

## Method: Modelling the FD-EF Relationship



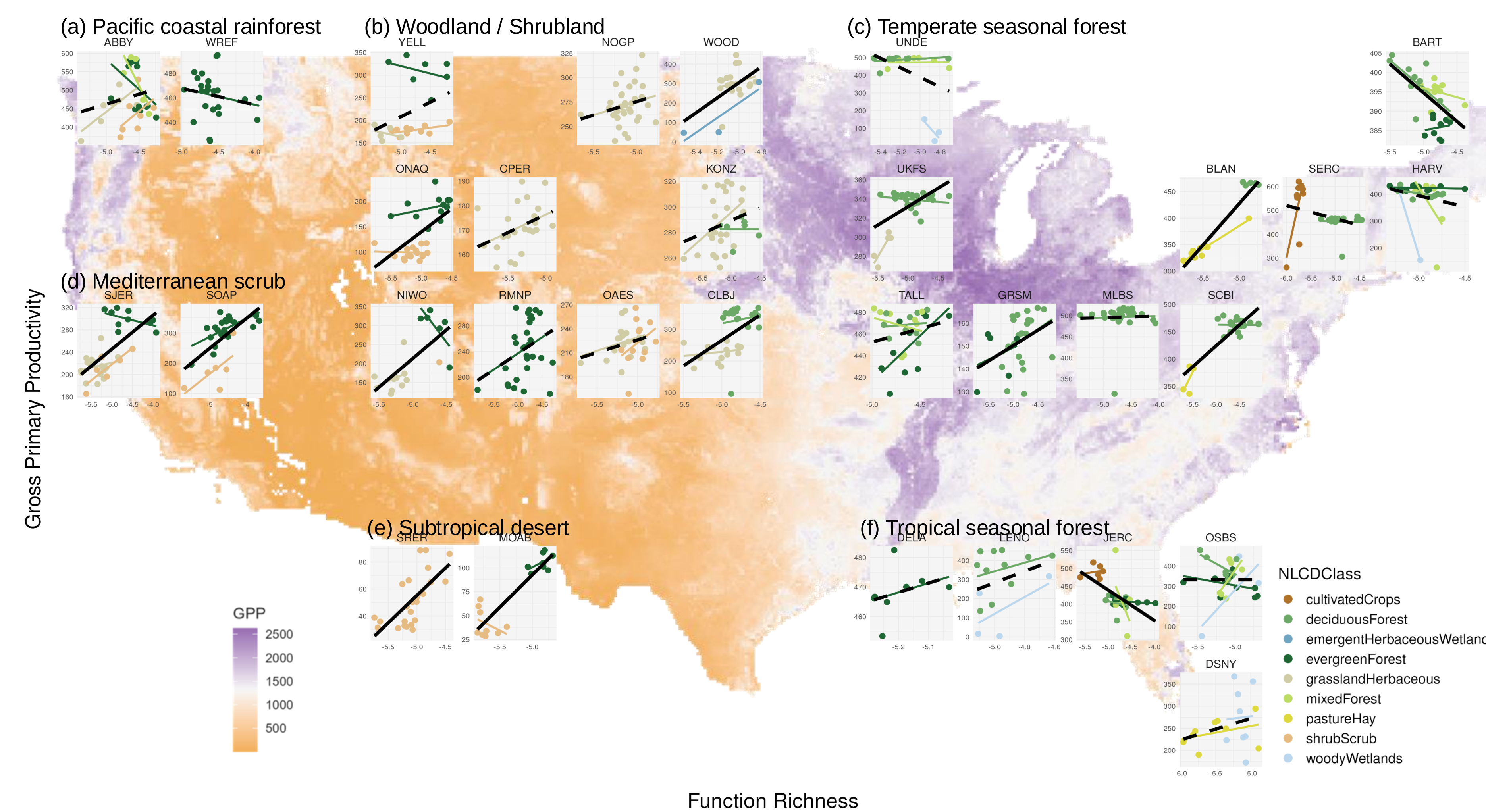
**Figure 1. Understanding the link between ecosystem functions and plant functional diversity.** This figure presents the methodological framework of our study. (a) Functional diversity metrics were derived from remotely sensed foliar traits, including chlorophyll, water content, carotenoids, etc. We then examined the relationship between functional diversity (FD) and various ecosystem functions, such as productivity and decomposition. (b) The 2-D illustration of the multi-dimensional FD calculation. In this study, we performed the FD calculation within a trait space.

## Results: Wall-to-Wall Map of Functional Diversity



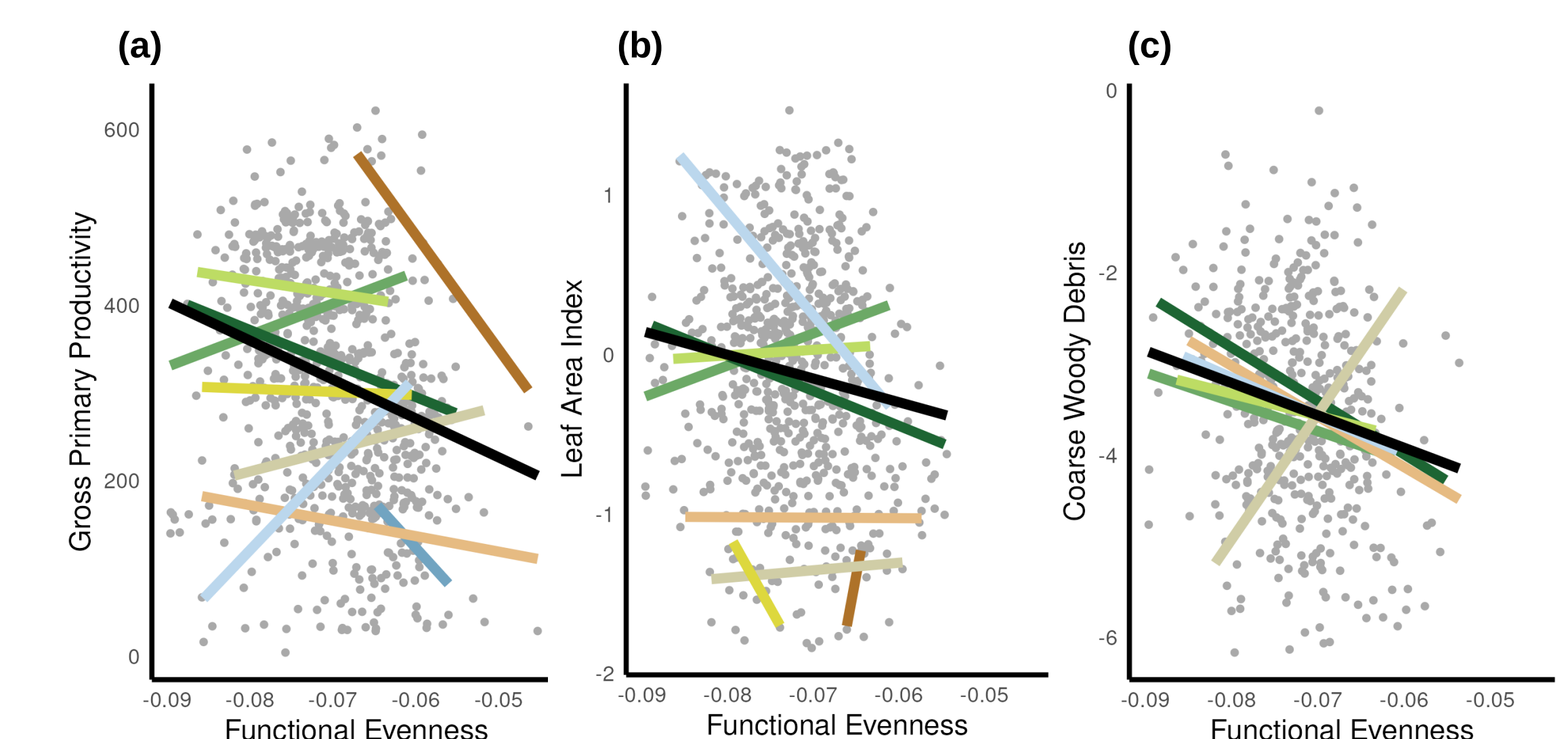
**Figure 2. Remotely sensed functional diversity map of site Harvard Forest (HARV).** Left panel shows functional richness and the right panel shows divergence. Diversity metrics have been standardized. Yellow boxes highlight some biodiversity hotspots: region (a) is coniferous forest; region (b) is recovering from recent disturbance.

## Results: Functional Richness Predicts Primary Productivity



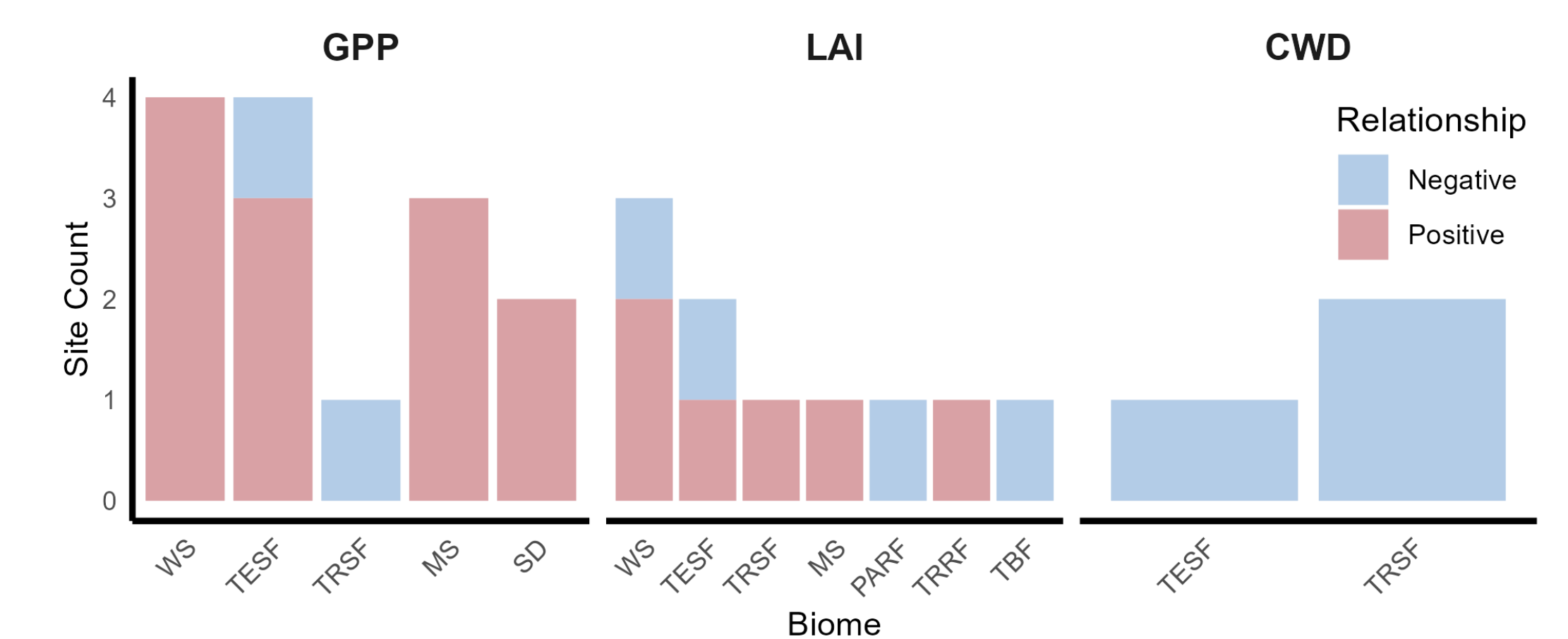
**Figure 3. GPP is positively correlated with functional richness across biomes.** Each subplot represents a NEON site, with the abbreviated site name (e.g., ABBY) as the subtitle. Sites that share similar geographical regions and climatic backgrounds are grouped in the same panel (biome). Black lines are the linear regression lines between GPP and functional richness for every site; solid lines represent significant FD-EF relationships ( $p < 0.05$ ), while dashed lines indicate weak relationships ( $p \geq 0.05$ ). For sites with negative or weak relationships, the relationships for individual vegetation types are shown by the colored lines. The basemap shows the GPP in July 2017 (from Landsat). The results of LAI show similar patterns across biomes.

## Results: Negative impact of Evenness



**Figure 4. Negative correlation between ecosystem function metrics and functional evenness.** (a) relationship between GPP and functional evenness ( $Corr_{overall} = -0.24$ ,  $p < 0.001$ ); (b) relationship between LAI and functional evenness ( $Corr_{overall} = -0.13$ ,  $p < 0.001$ ); (c) relationship between CWD and functional evenness ( $Corr_{overall} = -0.19$ ,  $p < 0.001$ ). The relationships for individual vegetation types are shown by the colored lines (legend can be found in Figure 3).

## Results: FD-EF varies across biomes



**Figure 5. The relationship between functional richness and ecosystem functions across biomes.** Y-axis shows the number of sites within each biome where the FD-EF relationship is significant ( $p < 0.05$ ). Red bars represent the number of sites with positive correlations, while blue represents negative correlations. The three sub-panels illustrate results for GPP, LAI, and CWD, respectively. For GPP and LAI, most biomes exhibit a positive FD-EF relationship, whereas CWD shows a predominantly negative relationship with FD across biomes. Abbreviations for biome name: PARF - Pacific coastal rainforest; TBF - Tundra or Boreal forest; TESF = Temperate seasonal forest; WS = Woodland or Shrubland; TRSF = Tropical seasonal forest; SD = Subtropical desert; TRRF = Tropical rainforest; MS = Mediterranean scrub.

## Conclusions

- Plant functional richness could be used as a good proxy of ecosystem productivity, allowing us to understand ecosystem functions at a large scale.
- Stronger FD-EF correlations were found in more diverse ecosystems, such as mixed and coniferous forests. By contrast, lower productivity and weaker FDEF relationships were found in wetland.
- Ecosystem decomposition, which is characterized by CWD, is negatively associated with functional richness.
- Our results can be adapted to satellite remote sensing, such as SBG and EMIT.

## Acknowledgements

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## References

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